

Problem formulation and motivation

Large Language Models (LLMs) have shown extensive ability for natural language processing tasks, but its use in clinical settings have to be approached with extra caution. This is primarily because LLMs have a habit of hallucinating, generating responses that are fluent but factually incorrect. In the medical domain, instances of LLM hallucination carry much heavier consequences than other fields.

Besides the issue of LLM hallucination, LLMs are trained on a fixed amount of knowledge. Knowledge in the medical field constantly evolves, so medical LLMs can quickly become outdated without regular retraining. Retrieval-Augmented Generation (RAG) is a common method used to address this issue. RAG enables an LLM to access relevant information from an external resource and uses that information as additional context at inference, reducing cases of hallucination and improving model performance without the need to constantly retrain the model.

When formulating the project, we decided to narrow the scope of the project to primary care. Primary care is the first and most frequent point of contact between the patient and healthcare system, covering a large range of medical topics. A medical question answering (QA) system can address the breadth of knowledge needed in primary care and holds the potential to alleviate the workload of those who work in primary care.

In our project, we will evaluate the impact of RAG on LLMs in the context of primary care. In addition to this, we will evaluate Retrieval-Augmented Fine-Tuning (RAFT), a system similar to RAG that trains the LLM with the context provided by RAG to enhance the use of the retrieved information.

Methodology and implementation

To build our retrieval pipeline, we used the MedRAG toolkit. For the corpus, Statpearls was chosen due to it being a lighter-weight corpus and extensive coverage of medical topics. The Statpearls corpus was chunked using MedRAG and encoded using MedCPT, a biomedical IR model that includes a pair of retriever and re-ranker closely integrated by contrastive learning.

After encoding, an FAISS index was built to enable efficient similarity search, resulting in faster RAG performance. With all these components completed, LLM with RAG would work as the following for a query: MedCPT encodes query → FAISS quickly retrieves StatPearls chunks → StatPearls chunks added as context for response generation.

BioMistral-7B was the choice of LLM to use as the basis of our experiment. It is a variant of Mistral-7B pre-trained on textual data from PubMed Central Open Access. We chose BioMistral-7B due to its domain relevance and competitive performance against other LLMs. BioMistral-7B was loaded in 4-bit quantization to be used in a Google Colab environment.

The dataset used for training and evaluation was MedMCQA. MedMCQA is an open source dataset containing multiple choice questions designed to address real-world medical entrance exam questions. To narrow the focus of the experiment to primary, subjects were filtered out via “subject_name” to include “Medicine”, “Pharmacology”, “Microbiology”, “Social & Preventive Medicine”, “ENT”, “Psychiatry”, “Orthopaedics”, and “Skin”. Once filtered, an 80/20 train/test split was used for the study.

Evaluation design

We built three models to evaluate given the components described above. Model 0 was the baseline model where BioMistral-7B underwent zero-shot evaluation with and without RAG. Model 1 was BioMistral-7B fine-tuned on MedMCQA without RAG context. Model 1 was evaluated with and without RAG. Model 2 was BioMistral-7B fine-tuned on MedMCQA with RAG context (RAFT). Model 2 was evaluated with RAG.

Our primary evaluation metric was accuracy (correct answers / total answers). In order to evaluate the training process we saved the midpoint model weights in order to later compare performance between different epochs. We also did some failure analysis in order to see failure patterns between RAG and non-RAG.

Results and analysis

As can be seen in Figure 1, on training we were interested to note that there was an early sudden drop off in training loss. This is likely due to the model getting a better sense of the

format rather than greater relation between the question and answers. While the model did continue to make modest improvements for each of the training sets, MedMCQA does not give clear reasons or logic for answers which could be the reason for the failure to continue to improve beyond the results shown in Figure 2.

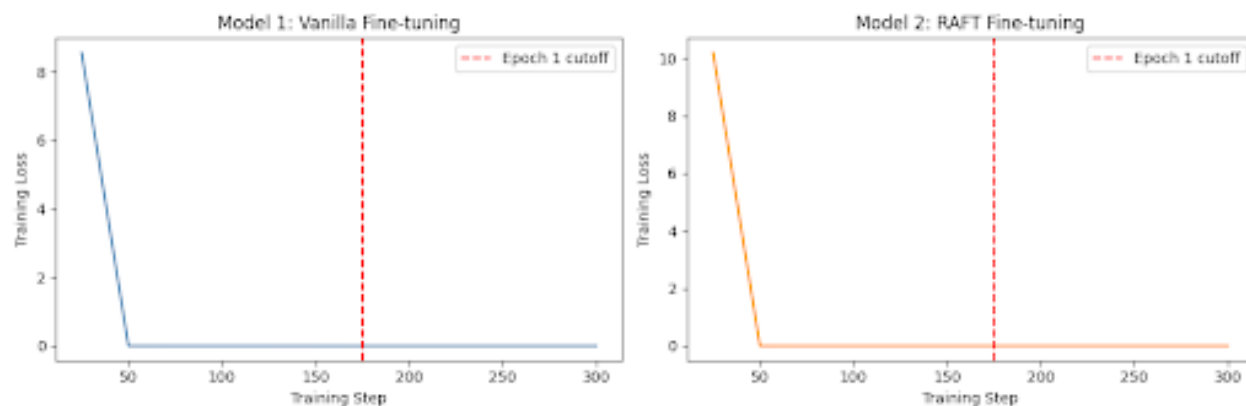


Figure 1: Loss rates over time

We were pleased to see that the model did improve with both the introduction of RAG and fine tuning. At baseline our model performed at 43.1% accuracy and this improved 3.5% with the addition of RAG. When fine tuned on model MedMCG, the non-RAG inference produced a 12% improvement to 54.4%. Adding RAG further improved the performance by 3.3% to 57.7%. Disappointingly including RAG during both training and inference did not improve performance above.

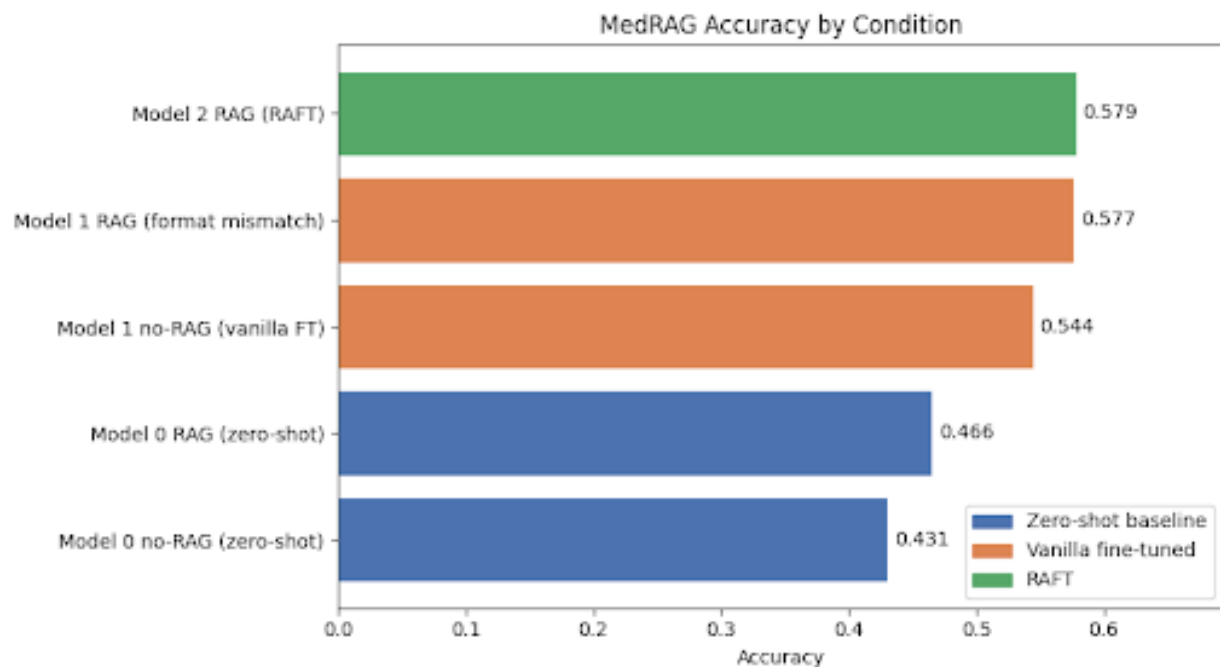


Figure 2: Results from the 5 arms

Because of the rapid drop in loss and subsequent stagnation, we decided to evaluate the model using only the weights from the first epoch. After this we reran the evaluation on a subset of the data to see if overtraining was readily apparent in the generalizability of the model. As can be seen in Figure 3, we had perhaps incorrectly chosen to limit Model 1 to 1 epoch but it seemed that the performance in Model 2 did peak in epoch 1. So interestingly, while the RAFT model was overtrained, the more heavily trained vanilla model would have performed better with additional training.

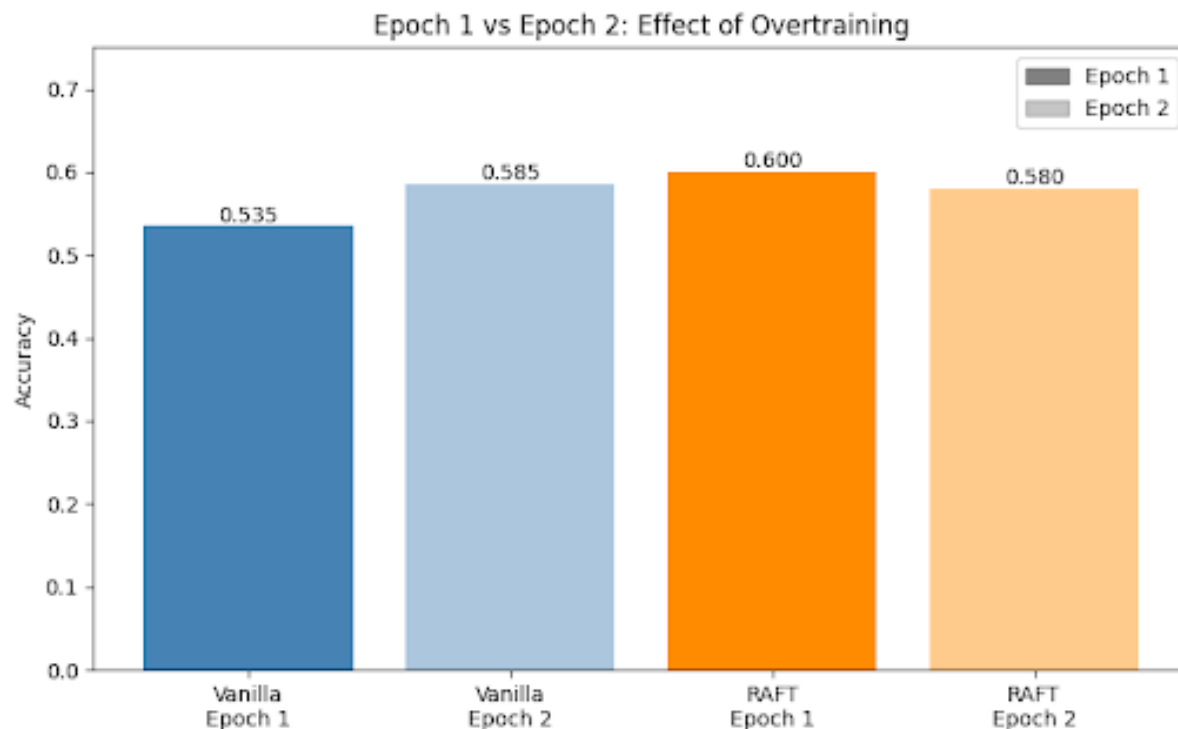


Figure 3: Overtraining analysis

We believe that potential improvements could result predominantly from improving the quality of the retrieval. On failure analysis, while we did see that the chunks being retrieved from the StatPearls corpus were indeed semantically appropriate, they did not always have specific relevance to the question being asked. For example, for a question asking for a categorization of a bat bite to a child's face, chunks from StatPearls about flea bits and human bites were pulled rather than bite categorization information. We suggest that this could have been significantly improved by running some chain of thought reasoning prior to retrieval to create a more suitable retrieval prompt and information that would help with answering the question rather than just being related in form. If we are looking for the evaluation of bites, we should be searching for chunks of that rather than chunks about bites in general. Furthermore, on RAFT, the training could have been done with the context of this updated prompt and could have potentially improved the performance of this step.

Chain of thought reasoning could have also been implemented at inference. At the moment, our model merely predicted the correct answer letter rather than having to reason in the context of the retrieved chunks. Having a prompt instructing the model to think deeply about why the options are correct or incorrect and consider whether the chunks apply to the given question

may also help avoid the retrieval potentially confusing the output with irrelevant but semantically convincing options.

A limitation we faced was in the hardware we had available. We initially had to move from PMC-LLaMA to BioMistral-7B because of recursion issues and subsequently kept running out of RAM to train the model. On upgrading to A100 GPUs on google colab, we were able to run the model, but still needed to take over an hour to train. This made iteration difficult.

Conclusion

Fine tuning a primary care medical question answering model and using StatPearls for RAG improved performance for the model. However, we believe incorporating chain of thought before retrieval, as well as at inference, would significantly improve the effectiveness. We would have also been interested to see if this greater context may have enabled the RAFT model to outperform the mismatch RAG plus vanilla model.

Team Contribution

We collaborated when deciding what structure we would use for the MedRAG. Andrew set up the initial Colab notebook with retrieval helpers, MedMCQA test and train datasets and basic evaluation. Nick chunked and embedded the StatPearls corpus, designed the experimental arms and trained and evaluated the model. We both contributed equally to the write up and presentation.